# LORENZO MASCIA

AI Engineer | GenAI & Agentic Systems | RAG + LLM Integration

Marigliano (NA), Italy • +39 389 542 3702 • lorenzo.mascia@gmail.com

howai.cloud • github.com/LorenzoMascia • huggingface.co/LorenzoMascia • linkedin.com/in/lorenzo-mascia

## SUMMARY

AI Engineer with 1+ year of experience developing AI driven solutions. Implemented LLM-driven dashboard generation and developed a modular correlation engine to support alarm analytics and event correlation. Open-source contributor (NL2Py, WASM RAG, LLM Distillery) with production-oriented focus on clean architecture and deployable AI systems.

## WORK EXPERIENCE

### ERICSSON — Product Developer (Full-time) — Aug 2025 – Present | Naples, Italy

- Built an AI-powered dashboard generation system (Grafana fork) enabling users to create complex monitoring dashboards via natural language.
- Designed a Go orchestration layer invoking a Python AI service that uses a RAG pipeline to extract schema/context from databases for LLM-based dashboard generation.
- Developed a React / Angular frontend supporting real-time dashboard rendering and interactive AI-driven configuration.

### ERICSSON — Integration Engineer (Internship) — Jan 2025 – Jul 2025 | Naples, Italy

- Developed an advanced Alarm Correlation Engine to detect related events and infer root causes in high-volume monitoring systems.
- Implemented a hybrid correlation approach combining DBSCAN clustering with weighted similarity metrics across temporal, spatial, semantic, and numerical dimensions.
- Integrated co-occurrence analysis on historical alarms to strengthen correlations through recurring patterns and improve robustness.
- Added rule-based correlation logic and RCA heuristics to increase interpretability and enable deterministic correlations alongside data-driven methods.

### PARTHENOPE UNIVERSITY OF NAPLES — Thesis Intern — Mar 2024 – Oct 2024 | Naples, Italy

- Built an ESP32 indoor tracking system for retail analytics (IR localization + real-time positioning).
- Implemented C firmware + customer movement clustering for behavioral insights.

## AI PROJECTS

### WASM RAG (Rust → WebAssembly)

Built a fully client-side RAG pipeline running entirely in the browser (Rust → WASM).
Implemented document parsing, chunking, embeddings, and vector similarity search without server infrastructure, enabling privacy-first and offline-friendly retrieval. (https://github.com/LorenzoMascia/wasm-rag)

### PII Anonymizer (LLM Fine-tuning + LoRA)

Released a 1B-parameter LLM fine-tuned for local PII anonymization before sending data to external AI services. Supports reversible anonymization via structured JSON token mapping. Published on Hugging Face with an interactive demo (https://github.com/LorenzoMascia/llm-distillery)

## EDUCATION

**BSc Management Engineering —** Univ. of Naples Parthenope (2020–2024)

**Exchange Program —** Universidad Politécnica de Madrid (2022–2023)

## SKILLS & CERTIFICATIONS

**SKILLS:** Python (PyTorch, Tensorflow, scikit-learn, Pandas), Go, Rust (WebAssembly), JavaScript, SQL, Git, Docker, CI/CD, REST APIs, Linux, Retrieval-Augmented Generation (RAG), Vector Databases (FAISS/Chroma), Prompt Engineering, LLM Integration (OpenAI), Clustering, Similarity Metrics (Cosine/TF-IDF)
**CERTIFICATIONS:** Microsoft AZ-900 • IBM RAG & Agentic AI • DeepLearning.AI ML Specialization
**LANGUAGES :** Italian (Native) , English (Fluent) , Spanish (Fluent)

## ADDITIONAL ACTIVITIES & ACHIEVEMENTS

**Kaggle**: Top 30% — Synthetic to Real Object Detection Challenge (dataset analysis, model training & evaluation).
**Open Source**: Contributor to Open Food Facts (GitHub PRs/issues, improving data quality & tooling).
**Technical Blog**: Author at **howai.cloud** — articles on LLM systems, RAG pipelines & applied ML engineering.